weakens the South-west monsoon.

Statement 4: CORRECT—During the south Asian summer monsoon, the TEJ induces secondary circulations that enhance convection over South India and nearby ocean.

The TEJ is the upper-level venting system for the strong southwest monsoon.

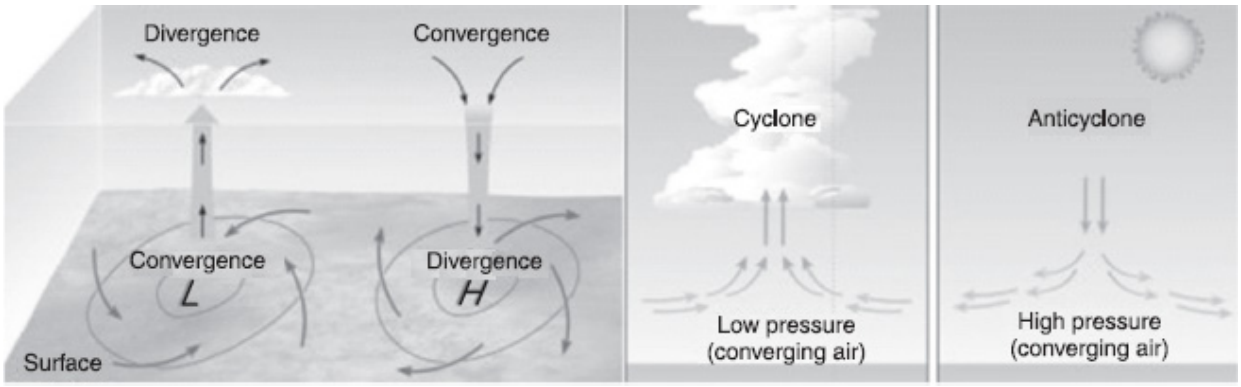
47. **(d) Occluded Front:**

Occlusion: Meteorology a process by which the cold front of a rotating low-pressure system catches up the warm front, so that the warm air between them is forced upwards.

Such a front is formed when a cold air mass overtakes a warm air mass and goes underneath it.

Frontolysis begin when warm sector diminishes and the cold air mass completely undertakes the warm sector on ground.

Thus, a long and backward swinging occluded front is formed which could be a warm front type or cold front type occlusion.



(a) **Conditions for the formation of air masses:**

- Source region should be extensive with gentle, divergent air circulation (slightly at high pressure).
- Areas with high pressure but little pressure difference or pressure gradient are ideal source regions.
- There are no major source regions in the mid-latitudes as these regions are dominated by cyclonic and other disturbances.

48. **Statement 1: CORRECT**—Relation between Absolute Humidity and Relative Humidity depends upon the temperature. With the change of air temperature, the capacity to retain moisture increases or decreases and the relative humidity is also affected.